

8.12 For every pair of variables (represented by nodes), there can either be a link connecting them, or not.

Total number of such pairs is given by: $(M - 1) + (M - 2) + \dots + 1$

$$= \sum_{i=1}^{M-1} i$$

$$= (M - 1)M/2$$

Since each of these pairs may or may not have a link connecting them, the number of distinct undirected graphs becomes:

$$= 2^{(M-1)M/2}$$

For M=3, the 8 possibilities are:

